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Title: Challenges and Opportunities for Artificial Intelligence in Auditing: Evidence from the Field

IEEE Citation

J. Kokina, S. Blanchette, T. H. Davenport, and D. Pachamanova, "Challenges and Opportunities for Artificial Intelligence in Auditing: Evidence from the Field," in *International Journal of Accounting Information Systems*, 2025, doi: 10.1016/j.accinf.2025.100734.

Summary:

This paper delves into the current state of artificial intelligence (AI) adoption in auditing, particularly within large public accounting firms, highlighting both the opportunities for efficiency gains and the significant challenges hindering broader implementation. Drawing from 22 semi-structured interviews with experienced audit professionals, the authors reconcile conflicting prior research some suggesting delayed AI uptake due to regulatory constraints and safeguards, others indicating widespread use by distinguishing between "simple AI" (e.g., tools for data extraction and optical character recognition) and "complex AI" (e.g., advanced machine learning for predictive analytics). The study reveals that simple AI is extensively integrated into routine audit tasks, improving accuracy and speed, while complex AI remains in developmental stages. Robotic Process Automation (RPA) is primarily applied to administrative workflows rather than core audit procedures, reflecting a cautious approach to automation in high-stakes environments.

The methodology centers on qualitative insights from the interviews, which uncover key barriers to AI adoption: lack of transparency and explainability (making it hard to justify AI decisions in audits), potential biases in algorithms that could skew results, data privacy concerns amid sensitive financial information, issues with robustness and reliability in varying scenarios, fears of auditors over-relying on AI and diminishing professional judgment, and the absence of clear regulatory or professional guidance. These challenges are exacerbated by the auditing profession's emphasis on accountability and compliance, where AI's "black box" nature poses risks. The authors also touch on related technologies like machine learning and natural language processing, noting their potential but limited current application in audits.

In the discussion, the paper proposes practical solutions inspired by the interviews and lessons from fields like healthcare and finance, such as developing explainable AI models, implementing bias audits, enhancing data governance frameworks, and creating industry-specific AI guidelines to build trust and reliability. Opportunities include leveraging AI for real-time risk assessment and fraud detection, potentially transforming auditing into a more proactive discipline. The authors conclude that while AI offers substantial benefits for audit quality and efficiency, overcoming these hurdles requires collaborative efforts from firms, regulators, and researchers. This work, part of a special issue on advanced audit technologies, underscores the need for balanced innovation to ensure AI enhances rather than undermines the auditing process, with confidential interview data supporting the findings.